

## CPR Esoft Quiz-4

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### Question #: 1

A 40-year-old man comes to the physician because of a 2-week history of nausea, vomiting, and abdominal pain. Physical examination shows no abnormalities. A gastric biopsy obtained from endoscopy shows a patchy erosion of the epithelial lining with mucosal infiltration by lymphocytes, macrophages, and plasma cells. A diagnosis of chronic peptic ulcer is made. Which of the following best describes this type of inflammation?

- A. Acute suppurative inflammation
  - B. Chronic granulomatous formation
  - C. Acute fibrinous inflammation
  - D. Acute serous inflammation
  - E. Chronic non-specific inflammation
- 

### Question #: 2

A 77-year-old man is brought to the emergency department from a nursing home because of a 1-day history of decreased consciousness. He was diagnosed with a perineal infection and started treatment with antibiotics 3 days ago. His blood pressure is 90/70 mmHg, pulse is 98/min and temperature is 104°F (40°C). Physical examination shows a febrile, comatose man with a soft mass in the perineal area. Laboratories studies show markedly elevated neutrophils. The physician suspects septicemia secondary to a perineal abscess. Which of the following mediators is most likely responsible for the fever observed in this patient?

- A. Histamine
  - B. Prostaglandins
  - C. Nitric oxide
  - D. Serotonin
  - E. Chemokines
- 

### Question #: 3

A 40-year-old man comes to the physician because of a 2-week history of nausea, fever, and chest pain. Physical examination shows an ill-looking, febrile man. Imaging studies show fluid in the pericardial cavity. Analysis of the fluid shows that it is rich in fibrinogen. Which of the following best describes this type of inflammation?

- A. Acute suppurative inflammation

- B. Acute fibrinous inflammation
  - C. Acute serous inflammation
  - D. Chronic non-specific inflammation
  - E. Chronic granulomatous formation
- 

**Question #: 4**

A 5-year-old boy is brought to the emergency department because of a 4-day history of dry cough and shortness of breath. He is a known asthmatic and has been hospitalized for exacerbation of this condition twice in the past 6 months. His blood pressure is 100/70 mm Hg, respirations are 26/min, and temperature is 36°C (96.8°F). Physical examination shows the use of accessory muscles and an expiratory wheeze. A diagnosis of acute exacerbation of bronchial asthma is made. Which of the following morphological findings best characterizes this condition?

- A. Increased destruction of elastic fibers
  - B. Thickening of the interalveolar septum
  - C. Constriction of the smooth muscles of the bronchioles
  - D. Squamous metaplasia of respiratory epithelia
  - E. Deficiency of surfactant production
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**Question #: 5**

A 35-year-old woman comes to the physician because of a 1-week history of a tender swelling in her right armpit. Physical examination shows enlarged right axillary lymph nodes. A biopsy is obtained from one of the enlarged organs for histopathological examination. Which of the following stains would most appropriately demonstrate the presence of the fibers in the stroma of this organ?

- A. Hematoxylin & Eosin
  - B. Trichrome
  - C. Sudan black
  - D. Giemsa
  - E. Silver nitrate
  - F. Oil red O
- 

**Question #: 6**

A 50-year-old man comes to the physician because of a 2-month history of poor appetite. He has a history of a 20-pack year of cigarette smoking. Physical examination shows scattered wheezes in both lung fields. A bronchial biopsy shows stratified squamous non-keratinized epithelium. Which of the following functions will most likely be affected as a result of this epithelial change?

- A. Inability to produce surfactant
  - B. Inability to adjust airway diameter
  - C. Inability to promote gaseous exchange
  - D. Inability to move mucus
  - E. Inability to secrete CC16
- 

**Question #: 7**

A 35-year-old man was brought to the emergency department because of sudden onset of severe abdominal pain following a car crash 20 minutes ago. Physical examination shows a distended abdomen. An ultrasound of the abdomen shows a ruptured spleen. Which of the following is most likely a characteristic feature of the red pulp of this organ?

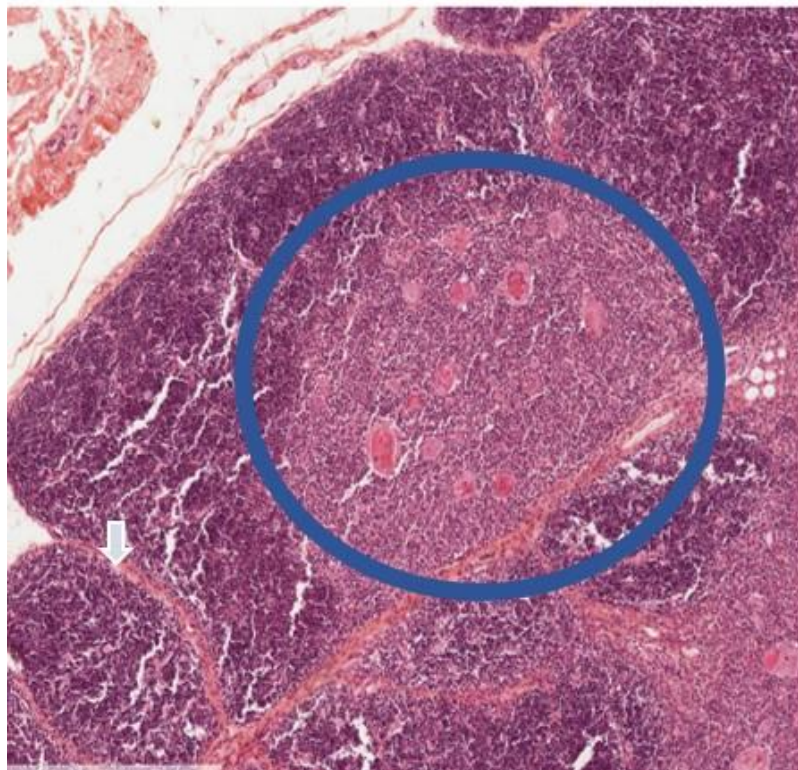
- A. Afferent lymphatics
  - B. Efferent lymphatics
  - C. Continuous capillaries
  - D. Fenestrated capillaries
  - E. Sinusoidal capillaries
- 

**Question #: 8**

A 3-year-old boy is brought to the physician by his father because of a 2-hour history of shortness of breath. He has a history of recurrent infections of the respiratory tract. He is suspected of suffering from a deficiency of protective lymphocytes which exposes him to both intracellular and extracellular infections. A stained section of the affected organ is shown in the attached photomicrograph. Which of the following best represents the cell that will most likely give rise to the eosinophilic corpuscles in the highlighted region of the affected organ?

- A. B-lymphocytes
- B. T-lymphocytes
- C. Reticular cells
- D. Epithelioreticular cells
- E. Plasma cells

**Attachment:**



*thymus .jpg*

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**Question #: 9**

A 49-year-old man comes to the physician because of a 2-month history of loss of appetite and unintentional weight loss. He has a 20-pack-year history of cigarette smoking. Physical examination shows a productive cough with sputum. A biopsy specimen of bronchial mucosa shows stratified squamous non-keratinized epithelium. Which of the following best describes the normal epithelium of this part of the airway?

- A. Stratified columnar
- B. Transitional
- C. Simple squamous
- D. Pseudostratified
- E. Simple cuboidal

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**Question #: 10**

A newborn boy is brought to the neonatal unit because of a 10-minute history of difficulty breathing and cyanosis. His mother has a history of hypothyroidism diagnosed 6 months ago. A diagnosis of neonatal respiratory distress syndrome is made. This condition is seen in infants with surfactant deficiency. Which of the following functions is most likely associated with this deficient compound?

- A. Secretion of a surface-active agent
  - B. Reduction of alveolar surface tension
  - C. Production of mucus
  - D. Formation of the blood-air barrier
  - E. Phagocytosis
- 

**Question #: 11**

A 59-year-old man comes to the physician for a follow up examination. He has a 10-year history of emphysema. Which of the following would most likely be expected of the position of the equal pressure point that develops during forced expiration?

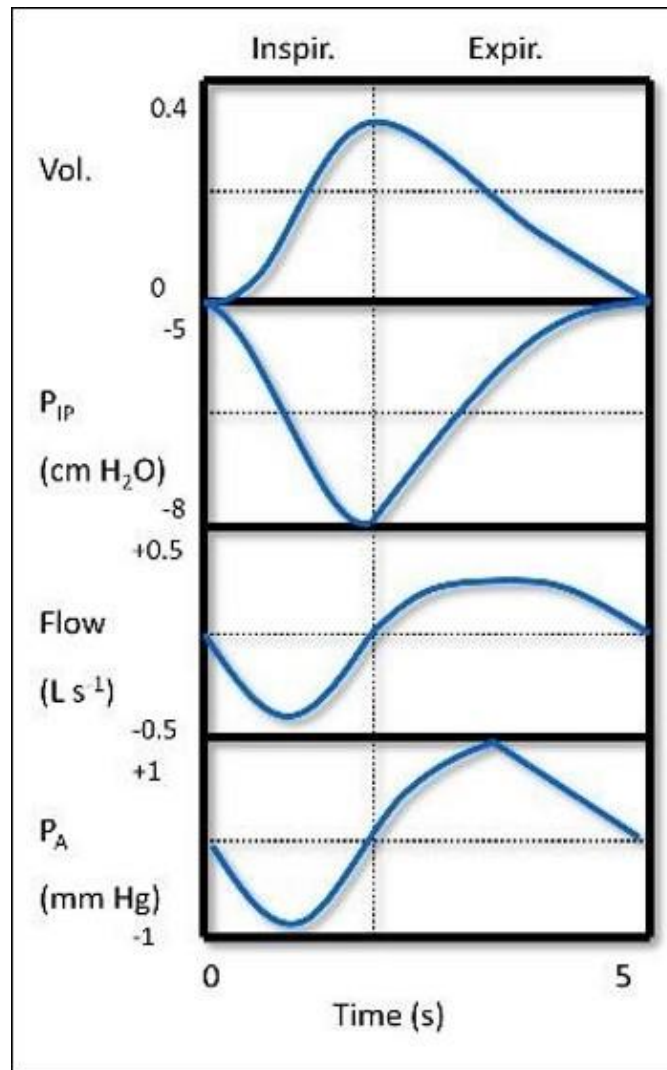
- A. It is not affected by static lung compliance
  - B. It moves proximally during a forced expiration
  - C. It is located distally from the mouth in a patient with emphysema
  - D. It is determined by the pulmonary blood flow
  - E. It is solely determined by the strength of the ventilatory accessory muscles
- 

**Question #: 12**

A 45-year-old woman comes to the physician because of a 2-day history of shortness of breath. She has no significant past medical history. Her temperature is 36.5°C (97.7°F), pulse is 90/min, respirations are 15/min, and blood pressure is 130/83 mm Hg. Her breathing cycle is evaluated, and the results are shown in the attached figure. Which of the following best explains a slight increase in alveolar pressure gradient during expiration compared to that in inspiration?

- A. Contraction of the intercostal muscles
- B. Relaxation of the diaphragm
- C. Decreased radial traction in the airways
- D. Decrease in alveolar volume
- E. Decreased thoracic cavity volume

**Attachment:**



Breathing cycle\_2.jpg

**Question #: 13**

A 56-year-old man is brought to the emergence department because of a 1-hour history of shortness of breath. He has a 15-year history of myasthenia gravis. His pulse is 88/min and respirations are 22/min with a decreased tidal volume. Physical examination shows weakness of muscles including muscles of breathing. Pulmonary function study shows a decreased change in thoracic volume during breathing cycle and shallow ventilation. Which of the following muscle weakness best explains shallow ventilation in this patient?

- A. Internal intercostal
- B. Diaphragm
- C. External intercostal
- D. Abdominal muscles
- E. Scalene and sternomastoid

**Question #: 14**

A 28-year-old healthy woman volunteers for a pulmonary study. She is asked to breathe normally through a spirometer and pressure measurement shows:

Alveolar pressure (PA) at the beginning of inspiration = 0 cm H<sub>2</sub>O,

Intrapleural pressure (PIP) at the beginning of inspiration = -4 cm H<sub>2</sub>O,

PA at the end of inspiration = 0 cm H<sub>2</sub>O,

PIP at the end of inspiration = -12 cm H<sub>2</sub>O,

Which of the following is the most likely transpulmonary pressure at the beginning of inspiration?

- A. -4 cm H<sub>2</sub>O
  - B. +4 cm H<sub>2</sub>O
  - C. -8 cm H<sub>2</sub>O
  - D. +8 cm H<sub>2</sub>O
  - E. -12 cm H<sub>2</sub>O
  - F. +12 cm H<sub>2</sub>O
- 

**Question #: 15**

A 34-year-old man is brought to the emergency department 30-minutes after he was involved in a vehicular collision. He sustained multiple rib injuries on the right side of his chest. His pulse is 104/min, blood pressure is 132/84 mm Hg, and respirations are 26/min with shallow tidal volume. Physical examination shows decreased respiratory movements, hyperresonant percussion notes and absent breath sounds in the right upper chest. An x-ray of the chest shows collapse of the right upper lobe of lung. He is diagnosed with right-sided pneumothorax. Measurements of pulmonary pressures on the right side shows an intrapleural pressure of 0 (zero) cm H<sub>2</sub>O at the end of expiration. Which of the following is the most likely cause of collapse of right upper lobe of the lung in this patient?

- A. A positive transpulmonary pressure of 10 cm H<sub>2</sub>O
  - B. Negative intrapleural pressure at the beginning of inspiration
  - C. A negative transpulmonary pressure
  - D. Zero transpulmonary pressure
  - E. Increased recoil of the chest wall.
- 

**Question #: 16**

A 53-year-old woman comes to the physician because of an 8-month history of progressive shortness of breath and fatigue. She does not smoke cigarettes or drink alcoholic beverages. Two years ago, she was diagnosed with lymphoma and had chemotherapy. Her respirations are 22/min with low tidal volume; other vital signs are within normal limits. Physical examination shows no abnormalities. A CT scan of the chest shows fibrosis of the lungs bilaterally. The pulmonary function studies show a flat lung compliance curve. Results of spirometry are shown:



Total lung capacity = 3.9 Liter,  
Functional residual capacity = 2.2 Liter,  
Intrapleural pressure at total lung capacity = -25 cm H<sub>2</sub>O,  
Intrapleural pressure at functional residual capacity = -5 cm H<sub>2</sub>O.  
Which of the following values is most likely close to the lung compliance in this patient?

- A. 85 L/cm H<sub>2</sub>O
  - B. 8.5 L/cm H<sub>2</sub>O
  - C. 0.085 L/cm H<sub>2</sub>O
  - D. 0.25 L/cm H<sub>2</sub>O
  - E. 0.17 L/cm H<sub>2</sub>O
- 

**Question #: 17**

A 56-year-old man comes to the physician because of a 6-month history of a dry cough and progressive shortness of breath. His respirations are 20/min and shallow; rest of the vital signs are within normal limits. Physical examination shows no abnormalities. Pulmonary function studies show a decreased FEV<sub>1</sub>, decreased FVC but an increased FEV<sub>1</sub>/FVC ratio. Which of the following mechanisms best explains these pulmonary study findings?

- A. Increased elastic recoil force
  - B. Decreased intrapleural pressure
  - C. Increased airway resistance
  - D. Increased compliance
  - E. Decreased chest wall recoil
- 

**Question #: 18**

A 59-year-old man comes to the physician because of a 4-week history of coughing and dyspnea. He had similar episodes three times in the last year. He has been smoking 3 packs of cigarettes a day for the past 37 years. His blood pressure is 140/85, pulse is 90/minute, and respirations are 24/minute. Physical examination shows use of accessory muscles for respiration with wheezing and crackles of the lung bases. An x-ray of the chest shows bilateral hyperinflated lungs and a flattened diaphragm. Which of the following is the most likely cause of the patient's presentation?

- A. Asymptomatic asthma
  - B. Chronic Bronchitis
  - C. Pneumonitis
  - D. Interstitial lung disease
  - E. Pulmonary Fibrosis
-



**Question #: 19**

A 68-year-old woman comes to the physician because of extreme dyspnea and a productive cough. She has a 60-pack year history of smoking. Physical examinations shows peripheral and central cyanosis and pursed lip breathing. She has been referred to the clinic for measurement of her FRC. Which of the following laboratory studies is most appropriate to measure her FRC?

- A. Bohr's method
  - B. Fowler single breath test
  - C. Flow-Volume spirometry
  - D. Helium dilution test
  - E. Plethysmography
  - F. Volume-Time spirometry
- 

**Question #: 20**

A 13-month-old boy is brought to the physician for follow-up examination because of failure to thrive and anemia. Genetic testing shows that one of the genes encoding beta-globin contains a pathogenic splice site mutation causing loss of the second exon. The second beta globin gene contains a promoter mutation which abolishes expression. Which of the following genetic principles is most likely highlighted in this patient?

- A. Loss of heterozygosity
  - B. Locus heterogeneity
  - C. Genetic linkage
  - D. Allelic heterogeneity
  - E. Anticipation
  - F. Haploinsufficiency
- 

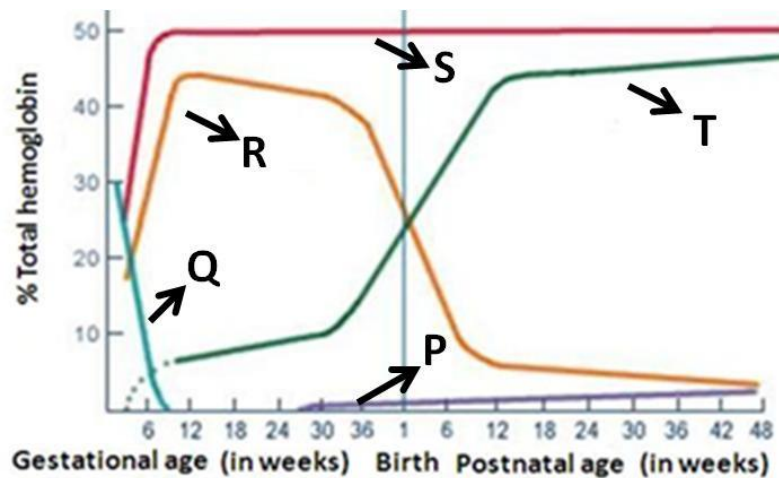
**Question #: 21**

A 3-year-old girl is brought to the physician by her parents because of a 2-day history of weakness and bone pain following a recent respiratory infection. Physical examination shows conjunctival pallor. Complete blood count shows anemia and hemoglobin electrophoresis shows an abnormal hemoglobin that has a lower mobility towards the anode than HbA. She is prescribed hydroxyurea. The attached figure shows the typical class switching of the globin gene loci. Which of the following lines in the figure represents the gene that is most likely to experience increased expression following treatment with this drug?

- A. The line labeled P
- B. The line labeled Q
- C. The line labeled R

- D. The line labeled S
- E. The line labeled T

Attachment:



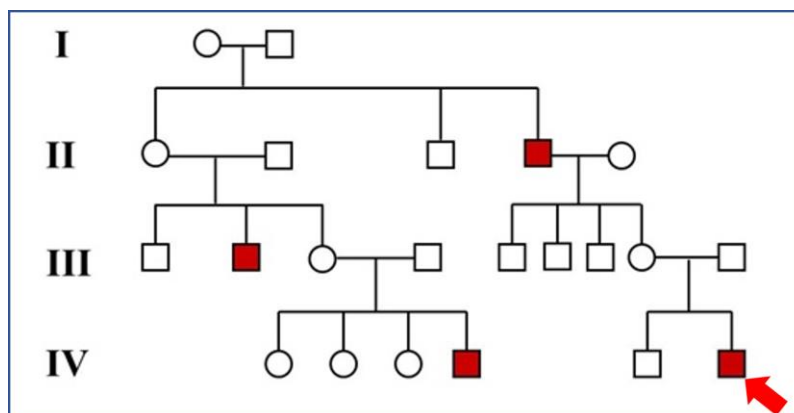
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#### Question #: 22

A 1-year-old boy is brought to the physician by his parents because of a 4-week history of repeated bruising. He has a 6-month history of painful joint swellings. Family history of the patient (IV-6) is shown. Laboratory studies show prolonged activated partial thromboplastin time (APTT). Prothrombin time (PT), platelet aggregation tests are within reference range. Which of the following disorders is most likely in this patient?

- A. Beta-thalassemia
- B. Cystic fibrosis
- C. Familial hypercholesterolemia
- D. Hemophilia A
- E. Sickle cell anemia
- F. Alpha thalassemia

Attachment:



pedigree.png

**Question #: 23**

A 30-year-old woman comes to the physician for a routine examination. She is in her third trimester. Ultrasound of the fetus shows herniation of part of the gut tube into the left hemithorax. A diagnosis of congenital diaphragmatic hernia is made. Which of the following is most likely caused by this condition?

- A. Respiratory distress syndrome
  - B. Hemothorax
  - C. Pneumothorax
  - D. Pulmonary hypoplasia
  - E. Eventration of the diaphragm
- 

**Question #: 24**

An autopsy on a 68-year-old man is performed, suspected of smoking is past. The lungs are removed, and large bronchopulmonary lymph nodes are observed. These nodes are most likely closely associated with which of the following structures?

- A. Carina
  - B. Hilum of the lung
  - C. Main bronchus
  - D. Lobar bronchi
  - E. Trachea
- 

**Question #: 25**

A 59-year-old woman comes to the physician because of a 1-month history of a sharp stabbing chest pain during exertion. Physical examination shows pleural rub during auscultation of the chest. A diagnosis of inflammation of the costal pleura is made. Which of the following nerves most likely carries the pain sensation from the affected structure to the CNS?

- A. Intercostals
  - B. Phrenic
  - C. Vagus
  - D. Thoracic splanchnic
  - E. Cardiopulmonary
- 

**Question #: 26**

A 20-year-old man is brought to the emergency department after a motor vehicle collision. His blood pressure is 65/45 mm Hg, pulse is 110/min and respiration are 24/min. Physical examination shows decreased breath sounds over the right lung fields. A diagnosis of tension pneumothorax is made. An emergency decompression is performed, and a chest tube is placed. In which of the following locations would the chest tube be most likely placed?

- A. At the top of the intercostal space
  - B. At the middle of the intercostal space
  - C. Above the superior border of the rib
  - D. Below the inferior border of the rib
- 

**Question #: 27**

A 1-day-old newborn boy is brought to the emergency department by his mother because of 1 hour history of not being able to breath. Physical examination shows rapid and labored breathing. Diagnosis of Infant Respiratory Distress Syndrome (IRDS) was made. What is most likely the cause of Respiratory Distress Syndrome?

- A. Pulmonary hypoplasia
  - B. Laryngeal atresia
  - C. Surfactant deficiency
  - D. Polyhydramnios
  - E. Tracheal atresia
- 

**Question #: 28**

An 18-month-old girl is brought to the physician by her parents because of a 6-hour history of inconsolable crying. She has a history of mild jaundice and intermittent episodes of pallor since infancy. Physical examination shows conjunctival pallor and mild scleral icterus. Laboratory studies show:

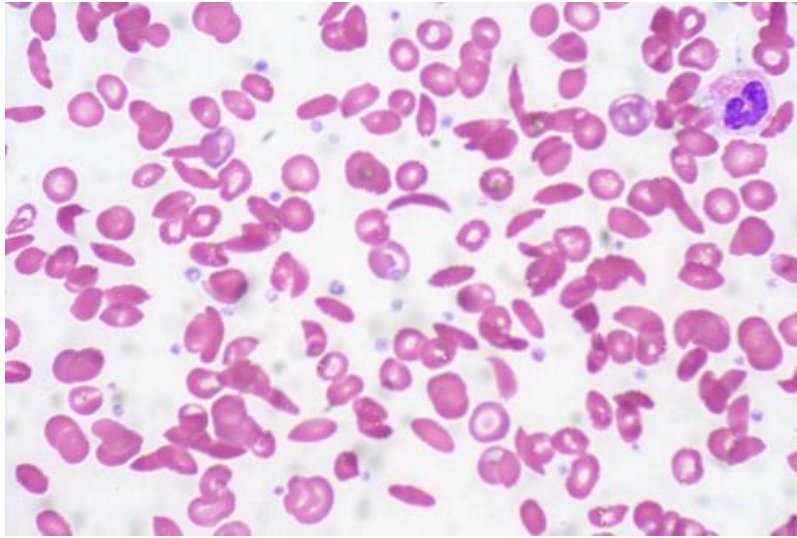
- Hemoglobin: 6.5 g/dL (low)
- Reticulocyte count: 12% (elevated)
- Total bilirubin: 3.2 mg/dL (elevated)

A peripheral blood smear is shown. Which of the following best describes the hemoglobin variant responsible for this patient's presentation?

- A. Higher affinity for oxygen than HbA
- B. Fewer number of negative charges than HbA
- C. Binds 2,3 BPG with lower affinity than HbA

- D. Contains iron in the ferric form
- E. Same primary structure as HbA
- F. Higher electrophoretic mobility towards the anode than HbA

Attachment:



92673.jpg

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**Question #: 29**

A 28-year-old man comes to the physician because of a 2-week history of fatigue, mild dizziness, and shortness of breath on exertion. He recently moved to a mountain village at an elevation of 3,500 meters. He says that he has been feeling progressively more tired over the past 2 weeks. His temperature is 37°C, pulse is 105/min, and blood pressure is 110/70 mmHg. Laboratory studies show mild hemoconcentration. His hemoglobin shows a sigmoidal oxygen dissociation curve with an increased P50. Which of the following best explains this alteration in hemoglobin function?

- A. Presence of fetal hemoglobin
- B. Increased pH
- C. Low PCO<sub>2</sub>
- D. High altitude acclimatization
- E. Hexokinase deficiency
- F. Decreased 2,3-Bisphosphoglycerate
- G. Decreased erythrocyte NADPH levels
- H. Increased erythrocyte glutathione levels

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**Question #: 30**

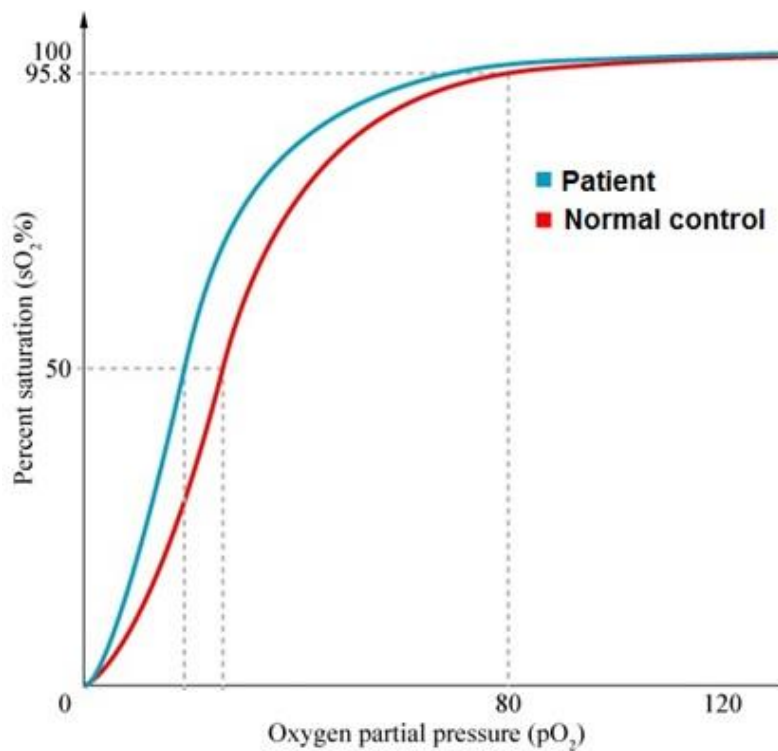
A 12-year-old boy is brought to the physician by his parents because of a 6-month history of fatigue and generalized weakness. Physical examination shows a pale complexion and conjunctival pallor. Laboratory studies show:

- Hemoglobin: 8.0 g/dL (low)
- Hematocrit: 28% (low)

The oxygen dissociation curve plotted from his hemoglobin is shown. Which of the following best explains his altered hemoglobin-oxygen binding affinity?

- A. Pyruvate kinase deficiency
- B. High altitude adaptation
- C. Hexokinase deficiency
- D. Carbon monoxide poisoning
- E. Sickle cell disease

**Attachment:**



170500.jpg